EDMS 10-100-2

SPECIFICATION

FOR

PIN-TYPE PORCELAIN INSULATOR

11 kV

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CONTENTS

1. SCOPE

This specification describes the minimum technical requirements for design, engineering, manufacturing, testing, inspection, packing, and performance of outdoor Pin-type porcelain insulators. The required insulators are intended to be installed in the overhead lines of the medium voltage distribution networks (12 kV) of the Egyptian Electricity Distribution Companies (EDCs).

2. APPLICABLE STANDARDS

Unless otherwise specified in this specification, the offered insulators should be designed, manufactured and tested according to the latest editions of the relevant applicable standards given in Table (1).

Table (1)

Standard No.	Description
IEC 60383-1	Insulators for overhead lines with a nominal voltage above 1000 V - Part 1: Ceramic or glass insulator units for a.c. systems - Definitions, test methods and acceptance criteria

3. ENVIRONMENTAL CONDITIONS

The electrical and mechanical properties of the required insulator should be guaranteed under the environmental conditions given in Table (2). Any differences in the guaranteed performance should be clearly set out in the offer.

Table (2)

Ambient temperature	Minimum mean daily: - 5°C Maximum mean daily: +45°C (50°C as option according to …EDC requirement)
Relative humidity	Minimum: 20%, Maximum: 100 %
Altitude	Up to 1000 m above sea-level
Pollution	Heavy (≥ 50 µS)
Average thunder storm	45 Days per annum.
Maximum annual rainfall	250 mm
Maximum wind speed	35 m/sec
Solar energy radiation	≥ 1100 W/m²

4. SYSTEM CHARACTERISTICS

The required insulator should have a safe electrical insulation behavior for the electrical system given in Table (3).

Table (3)

Nominal voltage	11 kV
Highest system voltage	12 kV
Power frequency	50 Hz
Number of phases	Three-phase
System earthing	Solidly earthed

5. TECHNICAL REQUIREMENTS

5.1 The Insulator:

– The insulator shall be made of best quality glazed porcelain free from flaws, cracks, voids, conducting impurities and chipped edges.

- The glaze shall be uniform, smooth, tough, brown in color and free from any kind of defects. The material of insulator shall be thoroughly vitrified so that the insulating property is independent upon the glaze.
- The porcelain shall not be directly engaged with metal. For fixing the pin to the insulator, a lead or copper thimble shall be cemented to the insulator by means of neat cement.
- The insulator shall be of one piece construction , the upper groove radius shall not be less than 12 mm, the insulator and its pin shall not be affected during service by the atmospheric conditions like salts, fumes, ozone, acids, alkaline, dust and sandy storms.

5.2 The Pin and Fittings:

- The pin shall be made of hot-dip galvanized steel. Each pin shall be complete with flat washer and two nuts of steel which shall be also hot-dip galvanized.
- The galvanization of pin, washer and nuts shall be in accordance with IEC standards.
- The pin shall have a shank of total length not less than 170 mm. The shank shall have an unscrewed part of length not more than 50 mm (see attached drawings). A drawing for the pin dimensions is given at the end of this specification.
- The insulator complete with its fitting shall not create excessive or localized corona at a voltage less than its rated voltage. The design of insulator and its fitting shall be such that to minimize local corona formation and discharges causing radio interferences.
- All insulator fittings shall have a factor of safety not less than 2.5 times the elastic limits of their materials.
- The electrical and mechanical characteristics for pin–type insulators shall be as given in Table (4).

Table (4)

1	Maximum system voltage (kV)	12
2	Operating voltage (kV)	11
3	Nominal Frequency (Hz)	50
4	Wet power frequency withstand voltage (kV)	≥ 45
5	Dry power frequency withstand voltage (kV)	≥ 80
6	Impulse 1.2 / 50 µs withstand voltage (kV)	≥ 125
7	Creep distance not less than (mm)	495
8	Minimum breaking load (kg)	900
9	15 minutes power withstand voltage at pollution of conductivity not less than 50 micro siemens (kV)	≥ 9
10	Power frequency puncture voltage (kV)	≥ 135

6. PACKING and MARKING

- The insulators should be packed in suitable wooden boxes to withstand shipment, transporting, handling and storing.
- Each insulator shall bear a permanent marking in accordance with IEC standard. The use of stickers/labels is not permitted.
- Each insulator shall be clearly and inedibility marked with the following:
  a) Mechanical and Electromechanical breakdown voltage
  b) Manufacture's name and trade mark
  c) Year of manufacture
  d) Name of manufacturing country
  e) Type and model of insulator

7. TESTING and INSPECTION

- All tests shall be carried out by and at the expense of the supplier in accordance to the latest relevant standard.

– Pin–type insulator shall be tested according to table 1 of IEC 60383-1 (page 71) for sample and routine tests. Type tests certificates from a neutral Laboratory shall be submitted with the offer.

– The insulators shall be tested for determination at sample determined according to IEC 60383-1 (clause 8.2). For sample tests, two samples are used E1 and E2. The sizes of these samples are indicated in Table 5.

Table (5)

Lot size (N)	Sample size E1	Sample size E2
N ≤ 300	Subject to agreement	
300 < N ≤ 2 000	4	3
2 000 < N ≤ 5 000	8	4
5 000 < N ≤ 10 000	12	6

– The minimum average thickness of galvanized (zinc coating) of metal parts shall be as follows:

- for pins : 85 micron
- for nuts and washers : 54 micron
- Sleeve should be made of zinc.

8. GUARANTEE

– The supplier guarantee the supplied materials against all defects arising out of faulty design or workmanship, or of defective material for a period of two years at least from date of delivery.

– Any defective parts of poor-quality or poor-assembly materials that are detected during the guarantee period should be replaced or repaired as soon as possible and free of charge.

9. SUBMITTALS

9.1 Submittals required with tender:

- The supplier shall complete and return one copy of the attached Data Schedule.
- Technical data, original catalogues and all necessary drawings including layout, dimensions shall be submitted with the offer.
- One specimen of the offered insulator shall be submitted with the offer for evaluation prior to issuance of the purchase order. The specimen should correspond in all respects with the material they intend to supply. Dimensional verification, indelible markings and finishing shall be checked.
- A valid approval certificate issued by Egyptian Electricity Holding Company.
- Copy of type test reports.
- List of deviations & clauses to which exception is taken (if any).

9.2 Submittals required following award of contract:

- Details of manufacturing and testing schedules
- Routine test reports
- The supplier should provide all the instructions of use/storing in (Arabic and English).

10. TECHNICAL DATA SCHEDULE

- The tenderer must fill in thoroughly the attached technical data schedule.
- Any offer does not accompanied with clear and complete technical data schedule shall be rejected.

Technical Data Schedule for Pin-type Insulator (12 kV)

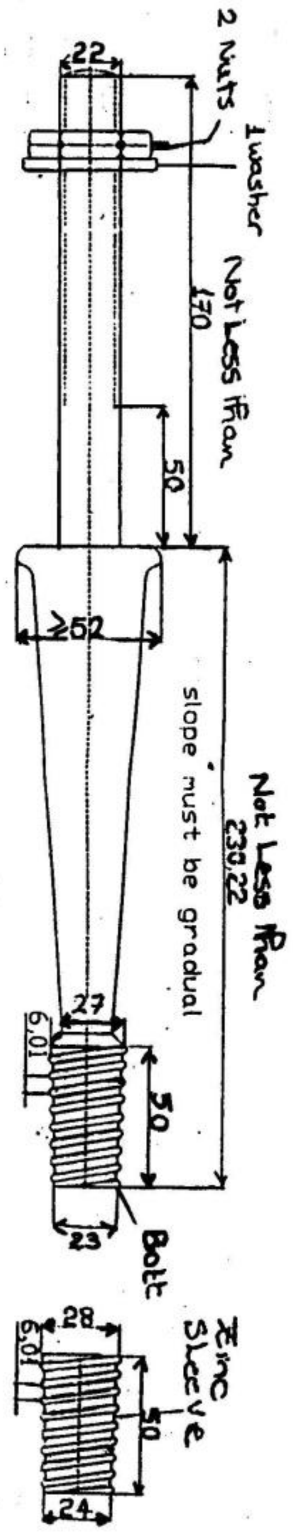
NO	ITEM	UNIT	Vendor Specifications
A. Dimensions, Weights and Particulars:			
1	Manufacture's name	-	
2	Type of insulator	-	
3	Greatest diameter of insulator	mm	
4	Overall length of insulator	mm	
5	Leakage distance not less than	mm	
6	Weight of insulator	kg	
7	Method of attachment of pin and insulator	-	
8	Number of pieces per box	pcs	
9	Marking as per clause 6	Yes/No	
10	Guarantee period	Year	
B. Mechanical Characteristics:			
1	Minimum breaking load	kg	
2	Critical load	kg	
3	Electro mechanical type test load	kg	
4	Maximum working load	kg	
C. Electrical Characteristics:			
1	Maximum wet power frequency withstand voltage	kV	
2	Maximum dry power frequency withstand voltage	kV	
3	Minimum wet power frequency flash over voltage	kV	
4	Minimum dry power frequency flash over voltage	kV	
5	Maximum dry lightning withstand voltage	kV	
6	Minimum dry lightning withstand voltage	kV	
7	Minimum power frequency puncture voltage	kV	
8	Maximum 15 minute power frequency withstand voltage at 50 micro siemens pollution	kV	
9	Minimum voltage at which corona appears	kV	

D. PIN:			
1	Length and diameter	mm	
2	Minimum breaking load	kg	
3	Minimum average thickness of galvanized for pin	micron	
4	Minimum average thickness of galvanized for nuts and washers	micron	

We guarantee the data given above for the offered insulator to …...EDC.

Signature: …………………………   Date: …………………….

Pin dimension of pin-type Insulator 12 kV



Note:

- The length L, diameter D, pitch, depth and slope of the pin threaded head must be equal to that of the threaded thimble fixed into the insulator.
- Minimum breaking load (failing load) of the pin shall not be less than 900 kg
- The pin shall be of hot - dip galvanized steel.